Event Set/Script Index	PHASE OF FLIGHT	I/E ACTIVITY	EXPECTED OUTCOME
0	**SETUP**	**I/E ACTIVITY**	**EXPECTED OUTCOME**
	SFO-LAX **SIM 1 FAA ID: 722** **PAN AM FAA ID: 1322** **SouthCoast 1 FAA ID: 267** **SouthCoast 2 FAA ID: 420** **Position:** Superbay Plot 40 **Route:** RWY10L SAHEY_ SUSEY EBAYE AVE SADDE_ RWY07R	**POSITION SET** **AIRPORT:** KSFO **RUNWAY:** 10L **POSITION:** RAMP 2 – EXEC (PAA) **Lat/Long:** N37*37:36/W122*22:42 Hdg 190* (South Coast) **AIRCRAFT SET** **APU:** Off **GROUND POWER:** EXT 1 **ZFW:** 238.7 **FUEL:** 31.0 **CG:** 24.3 **ENVIRONMENT SET** **WIND:** 150/10 **TEMP:** 11°C **ALTIMETER:** 30.11 in **RVR:** N/A **CLOUD CONDITIONS:** SCT 7500’ **VIZ:** 10.0sm **TURBULENCE:** Light 1 **RUNWAY CONDITION:** N/A **VISUAL CONTROL:** Current time of day	

1	PREFLIGHT TO ENGINE START :20 (0:20)	I/E ACTIVITY	EXPECTED OUTCOME
101	After briefing:	**Provide:** OFP/WX Paperwork, PF/PM role – Alternate from previous leg	Crew discussion TMP GC
102	*[D-ATIS in printed pkg]* **ATIS:** **KSFO DEP ATIS PAPA** xxxxZ Wind 150/10, VIS 10sm, SCT 075, T 11 /DP 04, QNH 30.11 IN, DEP RWY 10L, Advise on initial contact you have information **PAPA**	**TRIGGER: Close to departure time:** **MX informs the crew:** "Daily check has been accomplished. Aircraft Log has one deferral. We had to defer TR #4. Dispatch has been notified."	Crew discussion; MX, Dispatch coord CD WA EM WM PS
103	**IP Note:** *[Loadsheet in printed pkg]* *[FDP in printed pkg]* *[PDC in printed pkg]*	**TRIGGER: Crew seeks ATC clearance via voice or ACARS:** **San Francisco Delivery**: *"Giant 99 Heavy, San Francisco Clearance, cleared to Los Angeles International Airport via the SAHEY_ SUSEY transition, then as filed. Climb VIA the SID, expect Flight Level 390 one zero minutes after departure, departure on 135.1 Squawk 7225."* **San Francisco Delivery:** *"Giant 99 Heavy read-back correct, contact Ground when ready for taxi clearance."*	Crew discussion; MX, Dispatch coordination V TMR
104	**IP Note:** Select Pushback Left on IOS if applicable	**Ground Crew and ATC as necessary for an uncontrolled pushback**	The crew performs all cockpit preparations, checklists, engine starts TME TMP

2	ENGINE START THROUGH TAXI-OUT :10 (0:30)	I/E ACTIVITY	EXPECTED OUTCOME
201		**TRIGGER: Before Taxi Complete**	Observe Flaps set prior to A/C move, and ATC clearance to taxi 
202		**San Francisco Ground:** *"Giant 99 Heavy, taxi runway 10L via C, hold short runway 10L at R, you are number one for departure, tower 120.5, good-day."*	Crew Normal ops, SOP & Timing 
203	**IP Note:** Confirm Runway and Trim Verification Completed	**San Francisco Tower:** *"Giant 99 Heavy, Line up and wait runway 10L"*	Trim and runway verification is completed 

3	TAKEOFF THROUGH CLIMB :15 (0:45)	I/E ACTIVITY	EXPECTED OUTCOME
301		**San Francisco Tower:** *"Giant 99 Heavy, Wind 150/10, RNAV ORYAN, Runway 10L cleared takeoff"*	
302		**TRIGGER: Once Airborne** **San Francisco Tower:** *"Giant 99 Heavy contact NORCAL 135.1, good flight."*	Crew flies SID 
303	**IP Note:** Passing 5,000ft	**NORCAL Departure:** *"Giant 99 Heavy, radar contact, can you be above 14,000 in 4 minutes?"* **NORCAL Departure:** *"Giant 99 Heavy, roger"*	Crew altitude awareness & discipline 
304		**TRIGGER: Passing 10,000ft** **NORCAL Departure:** *"Giant 99 Heavy, contact Oakland Center 127.95"* **Oakland Center:** *"Giant 99 Heavy, radar contact, cleared direct EBAYE, Climb and maintain FL310, that will be your final today"*	Crew altitude awareness & discipline 

4	CRUISE TO DESCENT-DIVERT :15 (1:00)	I/E ACTIVITY	EXPECTED OUTCOME
401		**TRIGGER: Approaching EBAYE** **Oakland Center:** *"Giant 99 Heavy, contact LA Center 134.9"* **LA Center:** *"Giant 99 Heavy, radar contact"*	ATC Coordination WM
402	**IP Note:** QRH 2.4	**TRIGGER: After check-in** **Non-Normal: BLEED DUCT LEAK L**	Crew completes Non-Normal Checklist CM AM
403	**IP Note:** There are multiple scenarios with multiple options that may not have a script prepared for it. Do not pressure the crew in a specific direction or request contact with Maintenance or Dispatch. **Dispatch/MX/ATC as necessary Respond to PAN PAN or MAYDAY calls accordingly (make note if received for debrief)**	**Dispatch/MX and/or ATC as necessary, Dispatch requests continue to LAX if Crew elects otherwise.** ****Dispatch advises crew ACARS FDP Landing Distance Assessment and Live Calculation is unavailable**** **TRIGGER: Crew completes Dispatch coordination and/or QRH. Elects to continue to LAX.** **TRIGGER: After AVE** **LA Center:** *"Giant 99 Heavy, for traffic, descend and maintain FL290"*	Crew/Dispatch/ATC coordination (2 in 2 out) LF CD TMP
404	**IP Note:** Do not recommend or suggest using the QRH PI.22.11 or .12 chart for landing distance assessment. Make note if chart is not used for the de-brief	**TRIGGER: Descending through FL300** **LA Center:** *"Giant 99 Heavy, LA Center 130.25"* **LA Center:** *"Giant 99 Heavy, radar contact, cross SYMON at 12,000, LA altimeter 30.08"*	

5	DESCENT-DIVERT TO ARRIVAL :20 (1:20)	I/E ACTIVITY	EXPECTED OUTCOME
501	**IP Note:** Take note about flaps 25 vs 30 and autobrake selection planning for discussion during de-brief	**Trigger: Crew is in a decent through FL280** **LA Center:** "*Giant 99 Heavy, say airspeed?"* **LA Center:** "*Giant 99 Heavy, roger, maintain 320 knots in the transition"*	CM TME TMR
502	**IP Note:** Configure the IOS for LAX weather and landing runway 07R	**TRIGGER: Descending through FL250** **LA Center:** "*Giant 99 Heavy, cross SADDE at 250 knots, BAYST at 10,000"*	
503	*[D-ATIS in printed pkg]* **ATIS:** **KLAX ARR ATIS DELTA** xxxxZ Wind 150/28, VIS 1-1/2sm, SM BR, BKN 007, T 14 /DP 12, QNH 30.08 IN, ARR ILS RWY 07R, Advise on initial contact you have information **DELTA**	**TRIGGER: Descending through FL190** **LA Center:** "*Giant 99 Heavy, contact SOCAL approach 128.6, good day"*	Crew Briefing TMP

6	ARRIVAL TO APPROACH :10 (1:30)	I/E ACTIVITY	EXPECTED OUTCOME
601		**TRIGGER: Crew checks in to SOCAL** **SOCAL Approach:** "*Giant 99 Heavy, after SADEE direct TIMSE, ILS 07R, Los Angeles altimeter 30.08"*	
602		**TRIGGER: Descending through 11,000** **SOCAL Approach:** "*Giant 99 Heavy, descend and maintain 5000, speed 220 when reaching"* **TRIGGER: Past SADEE** **SOCAL Approach:** "*Giant 99 Heavy, Cross TIMSE at or above 2600, cleared ILS 07R, contact tower at FUMBL"*	

7	APPROACH THROUGH LANDING :10 (1:40)	I/E ACTIVITY	EXPECTED OUTCOME
701	**IP Note:** Headwind Component: 05kts Crosswind Component: 27kts VRef Additive: 05kts	**TRIGGER: Descending through 2500** Cobblestone Turbulence - Medium	PM assertiveness should the approach or missed approach become unstable. 
702	**IP Note:** **Make note of any stabilized approach criteria exceedances**	**TRIGGER: Crew contacts tower at FUMBL** **Los Angeles Tower:** *"Giant 99 Heavy, Los Angeles Tower, wind 150 at 28, runway 07R cleared to land"*	
703		**Los Angeles Tower:** *"Giant 99 Heavy, turn right when able, join ALPHA, ground point 75"*	
8	**TAXI-IN THROUGH POSTFLIGHT :05 (1:45)**	**I/E ACTIVITY**	**EXPECTED OUTCOME**
801	**IP Note:** Give the crew time to accomplish a "Flaps up after landing". Once complete, advise them we are complete with Leg 1	**Los Angeles Ground:** *"Giant 99 Heavy, stop on ALPHA"* **Advise crew:** Set the parking brake and leave the engines running. This is the end of leg 1. We will resume from this point.	

Event Set/Script Index	PHASE OF FLIGHT	I/E ACTIVITY	EXPECTED OUTCOME
0	**SETUP**	**I/E ACTIVITY**	**EXPECTED OUTCOME**
	LAX-ANC **Position:** Taxiway Alpha Departing Runway 25R **Route:** RWY25R MUELR_ RZS J88 PYE J143 ENI FOT KYLLE B453 MDO PTERS_	**POSITION SET** **AIRPORT:** KLAX **RUNWAY:** 25R **POSITION:** TAXIWAY ALPHA **AIRCRAFT SET** **APU:** Off **GROUND POWER:** N/A **ENGINES:** RUNNING **ZFW:** 242.1 **FUEL:** 61.0 [Set prior to break to allow for engines running] **CG:** 24.1 **ENVIRONMENT SET** **WIND:** 150/28 **TEMP:** 14°C **ALTIMETER:** 30.08 in **CLOUS CONDITIONS:** OVC 700’ **VIZ:** 1.5sm Smoke/Fog **RVR:** 6000 **TURBULENCE:** Cobblestone Light **RUNWAY CONDITION:** N/A **VISUAL CONTROL:** Current time of day	

1	PREFLIGHT TO ENGINE START :10 (0:10)	I/E ACTIVITY	EXPECTED OUTCOME
101	Prior to resuming	**Load cruise tailwinds. Recommend FL200 120 at 60 and FL300 120 at 90**	
102	After break:	**Provide:** OFP/WX Paperwork, PF/PM role – Alternate from previous leg	Crew discussion 
103	**ATIS:** **KLAX DEP ATIS Foxtrot** xxxxZ Wind 150/28, VIS -1/2sm SM BR, OVC 007, T 14 /DP 12, QNH 30.08 IN, DEP RWY 25R, Advise on initial contact you have information **Foxtrot** *[D-ATIS in printed pkg]*	**TRIGGER: Close to departure time:** **MX informs the crew:** *"Transit check has been accomplished. Aircraft Log has one deferral for Thrust Reverser #4, we have fixed the Bleed Duct Leak. Have a great flight"* **<u>Jumpseater</u>: Introduce yourself as a Company DH pilot that has a seat in the back and would like the right bunk if able to get some rest once airborne. Present your ID if requested.**	Crew discussion, MX, Dispatch coordination 
104	**IP Note:** *[Loadsheet in printed pkg]* *[FDP in printed pkg]* *[PDC in printed pkg]*	**TRIGGER: Crew seeks ATC clearance via voice or ACARS:** **Los Angeles Delivery**: *"Giant 411 Heavy, Los Angeles Clearance, cleared to Anchorage International Airport via the MUELR_, San Marcos transition, then as filed. Climb VIA the SID, expect Flight Level 340 one zero minutes after departure, departure on 125.2 Squawk 1705."*	Crew discussion; MX, Dispatch coordination 

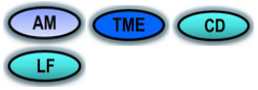
2	ENGINE START THROUGH TAXI-OUT :05 (0:15)	I/E ACTIVITY	EXPECTED OUTCOME
201	**IP Note**: Ground: 121.75	**Ground crew and ATC as necessary** **TRIGGER: Crew is complete with all briefings and cockpit preparations:** **Ground Crew informs the crew:** *"Ground to Cockpit, you have all engines running, wave-off/ salute has been received, flight controls have been checked, standing by for flaps and a Before Taxi Checklist*	The crew will perform a normal cockpit preparation, checklists, engine starts TMP
201		**TRIGGER: Before Taxi Complete**	Observe Flaps set prior to A/C move, and ATC clearance to taxi
202		**Los Angeles Ground:** *"Giant 411 Heavy, taxi runway 25R via A, F, cross runway 25L at F, hold short runway 25R at F, you are number one for departure, tower 120.95, good-day."*	Crew Normal operations SOP & Timing TME
203	**IP Note:** Verify Runway and Trim Verification Completed	**Los Angeles Tower:** *"Giant 411 Heavy, Line up and wait runway 25R"*	Trim and runway verification is completed WM IA TMR

3	TAKEOFF THROUGH CLIMB :20 (0:35)	I/E ACTIVITY	EXPECTED OUTCOME
301	**IP Note:** Tailwind Component: 05kts Crosswind Component: 27kts	**Los Angeles Tower:** *"Giant 411 Heavy, Wind 150 at 28, RNAV DOCKR, Runway 25R cleared takeoff"*	
302		**TRIGGER: Once airborne** **Los Angeles Tower:** "*Giant 411 Heavy, SOCAL Departure 125.2, good flight."*	Crew flies SID TME
303		**SOCAL Departure:** "*Giant 411 Heavy, radar contact, climb via the SID.*"	Crew altitude awareness & discipline TME V IA
304	**IP Note:** After A/P is engaged, if time is tight and in concert with tasks, use point directs to shorten the route	**TRIGGER: Passing 8,000ft** **SOCAL Departure:** "*Giant 411 Heavy, climb and maintain 15,000, expedite through 11,000 for crossing traffic"* [After CDU is programmed] **SOCAL Departure:** "*Giant 411 Heavy, climb and maintain FL190, maintain 280kts until FIXIT, Contact LA Center 127.8"*	Crew programs restrictions in the CDU AM
305		**LA Center:** "*Giant 411 Heavy, radar contact, climb and maintain FL340"*	Crew altitude awareness & discipline TME

4	CRUISE TO DESCENT-DIVERT :15 (:50)	I/E ACTIVITY	EXPECTED OUTCOME
401	**IP Note:** This should be a surprise for the crew, coincide with conversation or other distractors There are multiple directions that this scenario may take. This script is developed to accommodate a diversion to SFO or return to LAX. If alternate directions are requested, react accordingly.	**TRIGGER: Above FL300 but prior to FL340** **IOS: Initiate wake turbulence upset** **PAA: Malf Index>Flt Ctls>Unsched Stab Trim (NU) for only 5 seconds, then clear malfunction** **Sim 1: Malf Index>Flt Ctls>Excessive Roll (L or R)** **South Coast: Malf Index>Adv. Maneuvers>Excessive Pitch for only 5 seconds, then clear malfunction**	CM
402	**IP Note:** Once recovered, observe and react when the crew decides what to do.	**Give the crew a few moments to check on DH pilot or approach them "injured and needing care" after a delay. Strongly discourage a continuation to ANC.**	
402a	**IP Note:** **Divert to SFO** Do not coerce the crew in a certain direction. Take notes for discussion during the de-brief. **AFTER THIS SLIDE, SKIP TO SLIDE 501a**	**TRIGGER: Crew requests diversion to KSFO** **LA Center:** "*Giant 411 Heavy, copy diversion, contact Oakland Center 134.5. Good luck*"	Crew, Dispatch, ATC coordination (2 in 2 out) LF CD TMP
	BREAK	**BREAK**	**BREAK**
402b	**IP Note:** **Return to LAX** Do not coerce the crew in a certain direction. Take notes for discussion during the de-brief. **AFTER THIS SLIDE, SKIP TO SLIDE 501b**	**TRIGGER: Crew requests return to KLAX** **LA Center:** "*Giant 411 Heavy, direct FIM, descend and maintain 12,000, LA Altimeter 30.08*"	Crew, Dispatch, ATC coordination (2 in 2 out) LF CD TMP

5	DESCENT-DIVERT TO ARRIVAL :15 (1:05)	I/E ACTIVITY	EXPECTED OUTCOME
501a	**IP Note:** **SFO** **Configure IOS for SFO arrival weather and runway 28R** **Fail ILS or LOC and GS** [SFO *Arrival ATIS in printed pk*g] [SFO- *LDA in printed pk*g]	**(If WWAVS is behind the A/C, choose another waypoint further along the arrival)** **Oakland Center:** *"Giant 411 Heavy, direct WWAVS, descend via the SERFR_ arrival, San Francisco altimeter 30.11"* **KSFO Arrival ATIS:** *"San Francisco International Airport Arrival Information Whiskey at time xxxxZ, Wind 220 at 10kt Gust 25, visibility 10mi, FEW 075, Temperature 11 Dewpoint 04, QNH 30.11in, Arriving Runway 28R, Expect RNAV Z 28R. NOTAM ILS 28R is Out Of Service. Advise on initial contact you have information **Whiskey**."*	CM TME TMR
502a	**IP Note:** Take note about flaps 25 vs 30 and autobrake selection planning for discussion during de-brief	**TRIGGER: Descending through FL220** **Oakland Center:** *"Giant 411 Heavy, descend and maintain 7,000, maintain at least 280 knots as long as practical"*	Crew Briefing TMP
503a		**TRIGGER: Descending through 11,000** **Oakland Center:** *"Giant 411 Heavy, contact NORCAL Approach 134.5"* **Once complete, continue to slide <u>601a</u>**	
	BREAK	**BREAK**	**BREAK**

5	DESCENT-DIVERT TO ARRIVAL :15 (1:05)	I/E ACTIVITY	EXPECTED OUTCOME
501b	**IP Note:** **LAX** **Configure IOS for LAX arrival weather and runway 07R** **Fail ILS or LOC and GS** Headwind Component: 05kts Crosswind Component: 27kts VRef Additive: 05kts	**TRIGGER: Crew requests ATIS via ACARS or Voice** [*LAX ATIS in printed pkg*] **KLAX Arrival ATIS:** *"Los Angeles International ATIS information Delta, Wind 150 at 28, Visibility 1 and ½ mile Smoke Mist, Broken at 700, Temperature 14, Dewpoint 12, Altimeter 30.08, Arrivals expect RNAV Runway 07R, Advise on initial contact you have information Delta* [*LAX- LDA in printed pkg*]	Crew Briefing TMP
502b	**IP Note:** Take note about flaps 25 vs 30 and autobrake selection planning for discussion during de-brief	**TRIGGER: Descending through FL190** **LA Center:** "*Giant 411 Heavy, Contact SoCal Approach 128.6"*	
503b		**SOCAL Approach:** "*Giant 411 Heavy, after FIM expect RNAV Z 07R, Cross WAKER at 6000 and 210kts"* **Continue to slide 601b**	
	BREAK	**BREAK**	**BREAK**

6	ARRIVAL TO APPROACH :10 (1:15)	I/E ACTIVITY	EXPECTED OUTCOME
601a	**IP Note:** **SFO** **Select FAA Gust**	**TRIGGER: Crew calls NORCAL Approach** **NORCAL Approach:** *"Giant 411 Heavy, radar contact, Cross EDDY at 6000, cleared RNAV Z 28R Approach, Contact tower at AXMUL"* **Continue to slide 701a**	
	BREAK	**BREAK**	**BREAK**
601b	**IP Note:** **LAX**	**TRIGGER: Crossing FIM** **SOCAL Approach:** *"Giant 411 Heavy, Cross WAKER at 6000, cleared RNAV Z 07R, Contact tower at FUMBL"* **Continue to slide 701b**	

7	APPROACH THROUGH LANDING :05 (1:20)	I/E ACTIVITY	EXPECTED OUTCOME
701a	**IP Note:** **SFO** Monitor approach and note crossing restrictions, speed adherence and terrain awareness Headwind Component: 04kts Crosswind Component: 09kts Gust Additive: 15kts VRef Additive: 17kts	**TRIGGER: Crew contacts tower at AXMUL** **San Francisco Tower:** *"Giant 411 Heavy, Wind 220 at 10 gust 25, Runway 28 cleared to land"* **Continue to slide 801**	PM assertiveness should the approach or missed approach become unstable. A
	BREAK	**BREAK**	**BREAK**
701b	**IP Note:** **LAX** Monitor approach and note crossing restrictions, speed adherence and terrain awareness Headwind Component: 05kts Crosswind Component: 27kts VRef Additive: 05kts	**TRIGGER: Crew contacts tower at FUMBL** **Los Angeles Tower:** *"Giant 411 Heavy, Los Angeles Tower, wind 150 at 28, runway 07R cleared to land"* **Continue to slide 801**	PM assertiveness should the approach or missed approach become unstable. A

8	TAXI-IN THROUGH POSTFLIGHT :00 (1:20)	I/E ACTIVITY	EXPECTED OUTCOME
801	**IP Note:** Give the crew time to accomplish the situation or emergency. Once complete, advise them we are complete with the RLOFT and are going to accomplish **2 additional takeoff and landings each along with the SPOT training**. Configure IOS and setup using a sim reset. If any CBs are pulled to comply with DDG, reset them prior to landing currency. If there is seat support, only student landing currency is necessary. **Requirements:** • **Takeoff Departure Stall (390Kg)** • **Clean Configuration Stall** • **Landing Configuration Stall** • **KJFK-VOR 31L Approach** • **Visual Approach Pattern** • **Crosswind Landing (25+ kts)** • **Balked Landing** • **Landing Currency**	**IP Note:** **The RLOFT session is complete, the remainder of this session is SPOT training. Reposition to KJFK.** Complete the **JFK-VOR 31L approach**. Emphasis is on the visual approach segment along with stabilized approach criteria. Recommend winds 220/10-15 knots. For landing currency, **a visual pattern and manual flight is required**. This is to promote manual flying skills and pitch/power control as emphasized in FAA SAFO 13002. The instructor is encouraged to review pitch, power, energy management, and stabilized approach criteria. Discuss **Pilot Induced Oscillation (PIO), hard landings and strategies to mitigate these issues**. It is recommended that the visual approach landing is made with **25+ knots of crosswind** to emphasize the PIO and hard landing mitigation strategies. Baulked landings/low energy go-around versus full stop landings are acceptable and recommended at instructor discretion such that students leave with 3 takeoffs and 3 landings each (total). The aircraft must be within 50 feet of the runway, with the ability to land within the touchdown zone, and straddling the centerline, to count as a landing and subsequent takeoff. During these approaches, complete the SPOT training requirements of a **takeoff (390Kg), clean, and landing configuration stalls**. Recovery can be initiated **at buffet (natural) or stick shaker (artificial)**. Once complete, instruct them to taxi to park, and perform a shutdown and secure checklist. The words “Secure Checklist Complete” signify the conclusion of the simulator period.	